Lesson 2.9 Teaching Support Guide

Teach-tomorrow route

Print: current Guided Notes and Practice A for the class; Exit Tickets; optional Language Support and supported homework. Keep this guide and answer keys teacher-facing. Use the full-size model pages in the updated notes.

Phase	Planned minutes	Teaching move
Retrieve	5	Warm-Up items 1–4; note the missing prerequisite before launch.
Launch	7	Name the session target and invite a prediction from the conceptual task.
Model	10	Link a visual model to the worked example; rehearse the mathematical explanation.
Practice	18	Core: A #1, #2, #4, #6 and the error task. Choose remaining items from evidence.
Check	5	Exit Ticket; use the diagnostic page before assigning reteaching.

This is a planned 45-minute route, not a measured classroom duration. For a shorter lesson, keep the model and exit; reduce repeated practice. Form B is parallel practice or reassessment only when its items are unseen.

S1: Finding Mean Absolute Deviation (MAD) · I can find the mean absolute deviation of a data set and explain what it measures.

S2: Comparing Data Sets Using MAD · I can compare two data sets that share a mean by comparing their mean absolute deviations.

Spaced retrieval: revisit one item after 2 lessons and after approximately 1 week. Mix it with the next unit’s prerequisite check; do not announce the operation before students choose a method.

Session 1 · Finding Mean Absolute Deviation (MAD)

I can find the mean absolute deviation of a data set and explain what it measures. · 6.SP.B.5c

Observe and respond

Exit item	Expected evidence	Next prompt/model
1	a. 8 ÷ 4 = 2 b. 2 ÷ 4 = 0.5	Draw each distance as a segment from the mean. Add lengths, then divide by the number of observations.
2	Use distances 4, 1, 2, 3: sum 10; MAD = 10 ÷ 4 = 2.5	Draw each distance as a segment from the mean. Add lengths, then divide by the number of observations.
3	B — A typical value is about 4 from 72.	Ask the learner to show why B — A typical value is about 4 from 72. is correct and reject a specific distractor using a model.

Distinguish the error

Conceptual signal: Averages signed deviations or stops after adding the distances..

Response: Draw each distance as a segment from the mean. Add lengths, then divide by the number of observations.

Also check: Adding signed differences so they cancel to 0 instead of using distances.

Also check: Dividing the sum of distances by the mean instead of by the number of values.

If calculation alone is wrong, use Session 1 Prerequisite Practice A for the matching skill, then return to the model. If the interpretation is wrong, label the quantities before reteaching the operation. Independent re-entry requires a correct setup and a connected explanation.

Session 1 · Fresh check and scoring

Reserve this check until after revision; it is distinct from the exit ticket. Do not show this teacher page to learners beforehand.

Find the MAD of 3, 5, 7, 9 cm and interpret it.

Teacher solution: Mean 6; distances 3,1,1,3; MAD 2 cm, the average distance from the mean.

Score the evidence

Score	Evidence	Next action
3	Valid representation, correct result/units, connected explanation.	Continue; fade a support next time.
2	Correct setup and meaning with a calculation/notation slip, or an incomplete explanation.	Brief targeted feedback, then another unseen check.
1	Some relevant facts but a model or interpretation error.	Use the named model and the corresponding prerequisite page.
0	No interpretable evidence yet.	Oral/diagram response to separate access needs from mathematics.

Complete response anchor: Mean 5 cm; distances 3, 1, 1, 3 cm; total distance 8 cm; MAD 2 cm. Distances are nonnegative; using signed differences would cancel and would not measure typical distance.

Misconception response anchor: Averages signed deviations or stops after adding the distances.. Use draw each distance as a segment from the mean. Add lengths, then divide by the number of observations.

Language access: point/draw and label → oral rehearsal → relationship sentence → independent explanation. Offer the support needed for this task and fade it based on evidence. Suggested frame: The MAD is ___ because the total distance ___ is divided by ___. It describes ___.

Session 2 · Comparing Data Sets Using MAD

I can compare two data sets that share a mean by comparing their mean absolute deviations. · 6.SP.B.5c

Observe and respond

Exit item	Expected evidence	Next prompt/model
1	a. 8 ÷ 4 = 2 b. 20 ÷ 4 = 5	Keep the mean fixed, then compare every distance from it rather than recomputing only the centers.
2	Set M (MAD 1.5 < 7, so its values stay closer to 25)	Keep the mean fixed, then compare every distance from it rather than recomputing only the centers.
3	B — Their spreads may differ.	Ask the learner to show why B — Their spreads may differ. is correct and reject a specific distractor using a model.

Distinguish the error

Conceptual signal: Assumes equal means imply equal distributions or equal variability..

Response: Keep the mean fixed, then compare every distance from it rather than recomputing only the centers.

Also check: Thinking a larger mean means a larger MAD (center and spread are different measures).

Also check: Choosing the set with the larger MAD as the more consistent one.

If calculation alone is wrong, use Session 2 Prerequisite Practice A for the matching skill, then return to the model. If the interpretation is wrong, label the quantities before reteaching the operation. Independent re-entry requires a correct setup and a connected explanation.

Session 2 · Fresh check and scoring

Reserve this check until after revision; it is distinct from the exit ticket. Do not show this teacher page to learners beforehand.

C: 6,8,8,10 and D: 2,6,10,14 both have mean 8. Compare their MADs.

Teacher solution: C distances 2,0,0,2: MAD 1. D distances 6,2,2,6: MAD 4. C is more consistent around its mean.

Score the evidence

Score	Evidence	Next action
3	Valid representation, correct result/units, connected explanation.	Continue; fade a support next time.
2	Correct setup and meaning with a calculation/notation slip, or an incomplete explanation.	Brief targeted feedback, then another unseen check.
1	Some relevant facts but a model or interpretation error.	Use the named model and the corresponding prerequisite page.
0	No interpretable evidence yet.	Oral/diagram response to separate access needs from mathematics.

Complete response anchor: Both means are 5. A distances 1,0,0,1: MAD 0.5. B distances 4,2,2,4: MAD 3. A is more tightly clustered around its mean.

Misconception response anchor: Assumes equal means imply equal distributions or equal variability.. Use keep the mean fixed, then compare every distance from it rather than recomputing only the centers.

Language access: point/draw and label → oral rehearsal → relationship sentence → independent explanation. Offer the support needed for this task and fade it based on evidence. Suggested frame: Both sets have mean ___, but set ___ has a smaller MAD, so ___.